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1.0 OBJECTIVE:

1.1. The objective of this procedure is to define a process of deviation management at discovery GLP unit of Aurigene Pharmaceutical Services Limited.

2.0 SCOPE:

2.1. This SOP is applicable for deviations handling pertaining to study plan and SOP deviations at Aurigene Pharmaceutical Service Limited (APSL), Bollaram Road, Miyapur, and Hyderabad.

3.0 RESPONSIBILITY:

3.1. All Personnel who are responsible for performing activities/tasks should be aware of this SOP.

3.2. Study plan/Study specific SOP deviations

3.2.1. Unisite/multi-site study (test facility): Study personnel during the conduct of a GLP study shall report any deviations from applicable study plan, SOPs to respective Study Director/PI. All deviations shall be investigated to understand whether this deviation potentially impacts the quality and integrity of the GLP study as per the procedures explained in this SOP.

3.3. Study Personnel/SD/Contributing Scientist/Study pathologist are responsible for the assessment of root cause as applicable. Impact Analysis shall be done by Contributing Scientist/Study pathologist/Study Director. Study director shall ensure corrective and preventive actions are appropriately captured and adherence of the same. Study director are also responsible to provide appropriate justification for timeline extension where delay may be expected.

3.4. Multisite study (test site): Study personnel during the conduct of a delegated phase of GLP study shall report any deviations from applicable study plan, SOPs to the Principal Investigator. Study Personnel/Principal Investigator are responsible for the assessment of root cause as applicable. PI investigates to understand whether this deviation potentially impacts the quality and integrity of the delegated phase of the study and shall

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be approved by Principal investigator and acknowledged by Study Director. Principal investigator shall ensure corrective and preventive actions are taken and documented. To provide appropriate justification for timeline extension where delay may be expected.

3.5. Facility SOP deviations

3.5.1. All Personnel shall report deviations pertaining to facility to the respective Head-Function. Team leader/ Designee (person identified by respective Head-Function) in conjunction with Head-Function shall investigate the deviation and ensure corrective and preventive actions are taken and documented.

3.5.2. Team leader/ Designee (person identified by respective Head-Function) are responsible for root cause and impact analysis as applicable.

3.6. Head-Function/Department is responsible for ensuring implementation and compliance to the SOP in respective departments.

3.7. Head Quality Assurance or designee is responsible to review of deviations and ensure the compliance to this SOP.

3.8. Test Facility Management is responsible for ensuring overall compliance with this SOP

4.0 DEFINITION(S):

4.1. Corrections: Repair, rework, or adjustment relating to the disposition of an existing discrepancy/ non-conformity.

4.2. Deviation: An unintended departure from the defined procedures (e.g.Study plan/SOP) or an inadvertent change due to an oversight, equipment failure, etc with potential to impact on quality, safety or data integrity.

4.2.1. **Critical Deviation:** A Deviation where evidence exists that significant & unjustified departure(s) from Study plan, written procedure & the Principles of GLP has occurred resulting in

i. The test facility, or a part thereof, or a study is not in compliance with the Principles of GLP and/or

ii. A combination of several “Major” deviations across the basic GLP quality systems, and/or

iii. Deviation pertaining to data integrity related processes

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4.2.2. **Major Deviation:** A non-critical deviation where evidence exists that a significant departure(s) from protocol and written procedure has occurred that may not have developed into a critical issue, but it may lead to a facility, system or study being out of compliance, and/or where evidence exists of a failure of one of the basic GLP quality system elements.

4.2.3. **Minor Deviation:** Where evidence exists that a departure from the established guidelines and/or procedural requirements and/or Principles of GLP has occurred but does not affect the quality and integrity of the data.

4.3. **Study Director:** Study director is an individual responsible for the overall conduct of the non-clinical health and environment safety study. He/she represents the single point of study control with ultimate responsibility for the overall scientific conduct of the study. SD is usually the scientist responsible for study design and approval, as well as overseeing data collection, analysis and reporting. Study Director serves to assure that the scientific, administrative and regulatory aspects of the study are controlled.

4.4. **Principal Investigator:** An individual who, for a multisite study, acts on behalf of the Study Director and has defined responsibility for the delegated phase of the study. The SD's responsibilities for the overall conduct of the study cannot be delegated to the PI(s) which includes approval of the study plan and its amendments, approval of the final report and ensuring that all the applicable Principles of GLP area followed.

5.0 ABBREVIATION(S):

Abbreviation	Full Description
APSL	Aurigene Pharmaceutical Services Limited
BWD	Business Working days
CAPA	Corrective and Preventive Action
DC	Document control
GLP	Good Laboratory Practice
NA	Not Applicable
No.	Number
OECD	Organization for Economic Co-operation and Development
PI	Principal Investigator
QA	Quality Assurance
SD	Study Director

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Abbreviation	Full Description
SOP	Standard Operating Procedure
TFM	Test Facility Management

6.0 JSA

S.No	Potential Hazards	Recommended Safe Job Procedure
NA	NA	NA

7.0 CAUTIONS AND INTERFERENCES

Not Applicable

8.0 PROCEDURE:

8.1. Reporting and Categorization of Deviations

8.1.1. Deviation is an unintended change that occurs during the execution of activity using an applicable SOP/Study plan/method validation plan.

8.1.2. Responsible Personnel shall report deviations within **15 BWD** (Deviation reported date shall be considered as the date when the user requests to issue the deviation form/number from QA) from the date identified and close within **45 BWD** from the reporting date.

8.1.3. In circumstances where the investigation to derive root cause and impact assessment requires longer time due to dependency on the data analysis at the other phase or any other justifiable technical reasons then mail approval from Head-QA/designee shall be taken by the study director/PI as applicable.

8.1.4. Head-QA/designee shall assign an extension to a further **45 BWD** from the due date or completion of the experimental date (whichever is earlier) in order to facilitate the requirement. The same shall be updated in the Excel tracker i.e **FORM-GLP-GEN-0006.**

8.1.5. Responsible Personnel shall request QA to issue a Deviation number along with deviated SOP/Study plan and deviation details**.** QA shall issue deviation number and document details in an Excel tracker **FORM-GLP-GEN-0006.**

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8.1.6. Responsible person can download e-copy of the effective version of Deviation form **FORM-GLP-GEN-0005** from the Dochub(through "view native content"). In case of non-availability of the Dochub, the same can be sought from the document controller.

8.1.7. **Numbering System for Deviation**

DEV-XX-YY-NNN

Whereas,

DEV: Deviation.

XX : Departmental Code

NNN: Serial number (Starts from 001).

YY : Current Year

8.2. Study Plan/Study specific SOP deviations

8.2.1. **Deviation reporting**

8.2.1.1. Study personnel shall report deviations to SD/PI. Deviations pertaining to formulation preparation, dose formulation analysis, bio-analysis, toxicokinetic analysis, pathology or any other standalone activity shall be reported to respective contributing scientist/ study pathologist and /or SD/PI.

8.2.1.2. Personnel shall record details in the **FORM-GLP-GEN-0005,** such as Study plan with title and section number and SOP with version number, title and study number. Record section(s) deviated and briefly describe the deviation regarding what procedure was supposed to be followed and what was followed along with deviated duration (i.e., Record dates and time where applicable). If the deviation is associated with multiple SOPs which have interdependency; then all the SOP details shall be entered correspondingly.

8.2.1.3. Handover the document to contributing scientist or study director or PI as applicable.

8.2.1.4. Critical/major deviations which might affect the quality and end results will be intimated to sponsor.

8.2.2. **Root cause & impact analysis**

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8.2.2.1. Respective study personnel/contributing scientist/study pathologist/SD shall be responsible for investigating the proper root cause and Contributing Scientist/Study pathologist/Study Director shall be responsible for impact analysis for formulation preparation, dose formulation analysis, bio-analysis, toxicokinetic analysis, pathology or any other standalone report.

8.2.2.2. Root cause as well as impact analysis pertaining to any other study specific activity (other than mentioned in deviation form) shall be carried out by SD/PI.

8.2.2.3. Potential impact on the quality, integrity and scientific validity of data on the GLP study shall be assessed by:

a) Contributing scientist for standalone reports

b) Contributing scientist + study director for standalone reports (if required)

c) Study director/principal investigator as applicable

8.2.2.4. After completion of the assessment, study personnel will proceed with correction and the same shall be recorded in **FORM-GLP-GEN-0005** and Corrective and Preventive Actions shall be taken, as applicable. In a case where futuristic actions are required which triggers tracking of timeline, CAPA shall be initiated as per **SOP-GLP-GEN-0004.**

***Note**: CAPA form **(FORM-GLP-GEN-0007)** shall be initiated based on the requirement or as per the discretion of SD/PI in consultation with QA.*

8.2.2.5. Draft e-copy of the deviation shall be internally sent to all the stakeholders via e-mail. Once all the contents are finalized upon acceptance, the deviation shall be finalized and signed off post QA review. QA shall review and focus on the identified root cause and its impact on data integrity quality, corrections and appropriateness of CAPA.

8.2.2.6. SD shall approve the deviation, where APSL acts as test facility. In case of multi-site study where APSL acts as test site, PI shall approve the deviations and the same shall be acknowledged by study director.

8.2.2.7. Deviations that may potentially affect the integrity of the study shall be informed to SD and all such instances shall be also approved by SD and

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reported in the final report. Accordingly in case where APSL acts as test site, the deviations that may potentially affect the integrity of the study, PI shall inform SD and acknowledgment shall be sought.

8.2.3. Study Director/ Principal Investigators/ Contributing scientist shall implement the Proposed CAPA in conjunction with respective Head-Function/ Contributing scientist within the proposed timelines (as applicable).

8.2.4. Post implementation of the proposed CAPA, Study Director/ Principal Investigators/ Contributing scientist shall verify and inform QA for verification of the effectiveness of CAPA and Closure of CAPA.

8.2.5. QA personnel shall update the details in the Excel tracker **FORM-GLP-GEN-0006**.

8.2.6. **Reporting of Deviations**: **SOP-GLP-GEN-0023** shall be followed for the reporting of Deviations in the study report.

8.3. Facility SOP deviations

8.3.1. **Deviation reporting**

8.3.1.1. Personnel shall record details in the **FORM-GLP-GEN-0005**, such as SOP with version number and title. Record SOP section(s) deviated and briefly describe the deviation along with deviated duration (i.e., Record dates and time where applicable). If the deviation is associated with multiple SOPs which have interdependency; then all the SOP details shall be entered correspondingly. Record time where applicable.

8.3.1.2. Handover the document to respective Team leader or Head-function/Designee.

8.3.2. **Root cause analysis, Correction & CAPA**

8.3.2.1. Respective Team leader or Designee shall carry out an investigation to define the proper root cause and then assess impact analysis.

8.3.2.2. After completion of impact assessment, personnel shall proceed with correction and record the details in **FORM-GLP-GEN-0005** and if applicable. CAPA

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shall be documented. In case where futuristic actions are required which triggers tracking of timeline, CAPA shall be initiated as per **SOP-GLP-GEN-0004.**

Note: CAPA form shall be initiated based on the requirement or as per discretion of Team leader or Head-function/ Designee in consultation with QA.

8.3.2.3. Send the deviation reporting form to QA for review. QA shall review the entire form along with identified root cause and its impact on data integrity, quality, Corrections and appropriateness of CAPA.

8.3.2.4. Head-Function of respective departments shall approve the deviation.

8.3.2.5. Team leader or Designee shall implement the Proposed CAPA in conjunction with respective Head-Function within the proposed timelines (as applicable).

8.3.2.6. Post implementation of proposed CAPA, Head function shall verify and inform QA for verification effectiveness of CAPA as applicable.

8.4. Document Retention

8.4.1. After completion of all signatures, the original copy of deviation related to study shall be handed over to SD/PI and photocopy shall be retained at QA and authenticated by QA personnel. If the deviation is impacting many studies, then a photocopy shall be kept with all respective current studies and authenticated by respective study personnel. The original document shall be handed over to the respective SD or PI. SD/PI shall retain the original document in the respective study file.

8.4.2. After completion of all signatures, the original copy of the SOP deviation related to facility shall be handed over to QA and photocopy shall be retained at respective department and authenticated by respective department personnel.

8.4.3. QA shall archive deviation documents as per the procedure in **SOP-GLP-QA-0007.**

8.5. Trending of Deviations: To assess the type and number of deviations that occur in the facility, QA shall carry out trend analysis preferably every 6 months (depending upon the number of deviations) and present them to the test facility management to seek inputs and/or suggestions to improve quality systems.

8.6. Identification of Repeat Deviations:

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8.6.1. For the continuous improvement of quality systems at GLP, an assessment of Frequently occurring deviations shall be carried out wherein a specific methodology shall be followed as per **WI-GLP-GEN-0082** and based on the outcome appropriate actions shall be taken. This will help minimize the occurrence of repeated deviations and in turn provide sustainable GLP quality system.

9.0 REFERENCE(S):

Document No.	Title of the Document
NA	Quality Management systems- requirements ANSI/ISO/ASQ Q9001-2000
SOP-GLP-QA-0007	Submission of Documents maintained by Quality Assurance to Archives
SOP-GLP-GEN-0023	Preparation, review and Finalization of Study report and its Amendments
SOP-GLP-GEN-0004	CAPA Management
OECD Document number 24	Quality Improvement tools and GLP, July 2022

Note: refer current version of above-mentioned documents

10.0 ANNEXURE(S):

Document No.	Title of the Document
WI-GLP-GEN-0082	Identification of Repeat Deviations and mechanism for addressal
FORM-GLP-GEN-0005	Deviation reporting Form*
FORM-GLP-GEN-0006	Deviation Tracker#
FORM-GLP-GEN-0007	CAPA Form*

#Maintained as an Excel tracker

*Editable forms

Note: refer current version of above-mentioned documents

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11.0 CHANGE HISTORY

Quality Record No	Brief details of changes
CC1000093660	1)To Define the methodology for identification of Repeat Deviations and the mechanism to address in section 8.6 2) To define immediate Corrective and Preventive actions in the deviation form in section 8.2.2.4 & 8.3.3.2 and minor changes in the Form design 3)To give provision for extension of deviation in case where investigation, root cause and impact analysis require more time due to technical issues or dependency in section 8.1.3 4)To update the SOP template in order to meet current SOP requirements 5)To provide clarity on Reporting of Study specific deviation in Section 8.2.6 Editorial changes for better clarity

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Intention	Employee ID	User Name	Date (IST) DD/MM/YYYY
Prepared By	P00068975	G Bharath Kumar.	02/02/2023 10:28:34
Peer Reviewed By	P00068936	V Vijayabalaji.	02/02/2023 12:38:09
QA Reviewed By	P00070211	Nishant Christian.	02/02/2023 14:25:16
Approved By Test Facility Management	P00068654	D Ravi Krishna Babu.	02/02/2023 15:35:00

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1.0 OBJECTIVE:

1.1. The objective of this standard operating procedure is to give a brief description of archives and its maintenance procedures.

2.0 SCOPE:

2.1. This SOP is applicable to describe a detail description and maintenance of Archives (Documentation, Specimen, Test/Reference Item & Electronic Archives**)** at Aurigene Pharmaceutical Services Limited, Bollaram Road, Miyapur, Hyderabad, Telangana, India.

3.0 RESPONSIBILITY:

3.1. Archives staff shall be responsible for following the procedure described in the SOP and are responsible for maintaining logbooks and databases related to archives as per the procedure described in this SOP.

3.2. Archives staff shall be responsible to ensure pest control activities are carried out as per schedule in conjunction with SH&E department.

3.3. Archives staff shall be responsible to ensure EU related activities are carried out as per schedule in conjunction with EU department.

3.4. QA, SH&E and EU personnel shall be aware of this SOP.

3.5. QA personnel should be aware of this SOP.

3.6. Test Facility Management and/or Deputy shall be responsible to ensure overall compliance to this SOP.

4.0 DEFINITION(S):

4.1. **Archive:** A designated area or facility (e.g. cabinet, room, building or computerized system) for the secure storage and retention of records and materials.

4.2. **Archivist and Deputy:** An individual designated by test facility or test site management to be responsible for the management of the archive, i.e. for the operations and procedures for archiving.

4.3. **Archive Staff:** Archivist and the individuals who work under the supervision of the archivist and who are responsible for the routine archive operations.

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4.4. **Documentation Archives:** It is a site of archiving documents (paper copies)

4.5. **Specimen Archives:** It is a site of storage for specimen tissues, block and slides.

4.6. **Electronic Archives:** It is a site of storage of CD/DVD using a fire-proof cabinet. Desiccant silica gel bag is placed in the cabinet for absorbing the moisture content.

4.7. **Test/Reference item Archives:** It is a designated area or facility of archiving test/reference item.

5.0 ABBREVIATION(S):

Abbreviation	Full Description
APSL	Aurigene Pharmaceutical services Limited
EU	Engineering & Utility
e.g.	Example given
GLP	Good Laboratory Practice
IT	Information Technology
i.e.	That is
NA	Not Applicable
NBF	Neutral Buffered Formalin
No.	Number
PPE	Personal Protective Equipment
QA	Quality Assurance
RT	Room Temperature
SH&E	Safety Health & Environment
%	Percentage
˚C	Degree Centigrade

6.0 JSA

S. No.	Potential Hazards	Recommended safe job procedure
1	a. Test item exposure during weighment, handling, transfer may Cause Cancer, heritable genetic damage, very toxic by inhalation, in contact with skin and if Swallowed	a. Use the recommended PPE. Ensure adequate ventilation. Only authorized and trained people should carry out this job. Avoid contact with skin and eyes. Avoid formation of dust aerosols. Wash hands thoroughly after handling the test item
2	b. Wear recommended PPE	b. Wear recommended PPE and carry

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	and carry out the job using a fume hood.	out the job using a fume hood.
3	c. While transferring formaldehyde into the primary container ensure a secondary container is used to avoid spillage.	c. While transferring formaldehyde into the primary container ensure a secondary container is used to avoid spillage.

7.0 CAUTIONS AND INTERFERENCES

7.1. Wear appropriate PPE while handling specimens and test/reference items.

7.2. Be aware of all safe operating procedures to be followed in archives.

7.3. Smoking, eating or drinking are strictly restricted in archives. Smoking is a fire hazard, while food and drink can damage the archives both directly and indirectly through the encouragement of rats and other pests.

7.4. Wash hands before and after handling of specimens and 10 % NBF. Wash hands before eating. Avoid contact with eyes, skin, and clothing. Keep container tightly closed. Do not ingest or inhale.

7.5. Staff shall be trained in the safe techniques for lifting weights, handling of records and materials.

8.0 PROCEDURE:

8.1. Archive Facility

The designated archive facilities available at APSL are as follows;

a) Documentation Archives

b) Specimen Archives

c) Test/Reference Item Archives

d) Electronic Archives

8.2. General Description of Archives

8.2.1 The archive facility has been constructed to withstand the elements of local weather. The facility is also provided with back-up electrical power supply. The archive facility

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is suitably designed and constructed to accommodate archived records and materials.

8.2.2 Access to keys and swipe card (e-access) is provided to designated staff and strictly restricted. This helps to prevent any unauthorized entry to the archived records and materials.

8.2.3 Follow the entry/exit procedure of archives as per **SOP-GLP-AC-0004** (Access, Entry & Exit Procedures in GLP Archives).

8.2.4 Laptop is available in archives to facilitate the archival activities.

8.2.5 The shelves used to retain records and materials are non-combustible. Fire-proof lightening system is provided to minimize the risk of electric short circuit.

8.2.6 Aurigene facility is provided with fire-fighting equipment near to the repositories and an automated fire suppression system to minimize the risk of damage.

8.2.7 Smoke detectors are installed in all archives and performance verification shall be performed as per **SOP-GLP-EU-0010** (Periodic Verification of Smoke detection system).

8.3. Documentation Archives

8.3.1 Documentation archives are separate from the specimen archives.

8.3.2 The entry to documentation archives is secured by a strong door with a lock & key and access control system.

8.3.3 The records storage space in documentation archives are identified for study specific and facility records, with 9 racks numbered from R1 to R9 each. Each rack contains 7 rows and 3 columns numbered from A to G and 1 to 3 respectively.

8.3.4 An AC is installed in documentation archives to maintain the controlled room temperature.

8.4. Specimen Archives

8.4.1 The entry to specimen archives is secured by a strong door with a lock & key and access control system.

8.4.2 The storage space is divided into 8 Racks R1 to R8. Each rack contains 10 rows (shelves) identified as A to J and in each row 6 tubs can accommodate.

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8.4.3 Along with racks, 5 cabinets are available in specimen archives to store the tissue blocks and slides exclusively. Cabinet 1 & 3 are allotted store Slides and Cabinet 2, 4, 5 are allotted store blocks.

8.4.4 Specimen archives is connected with AHU to facilitate air circulation and to avoid exposure to formalin fumes

8.4.5 Eye wash bottle; PPE like lab coat, chemical resistant protective gloves, nose mask and chemical safety goggles shall be maintained in specimen archives for safe handling of specimens.

8.5. Test Item/Reference item Archives

8.5.1 Test/Reference item and electronic archives are within the Toxicology building. Access control system is installed at the main gate of the building.

8.5.2 Entry to test item/reference item and electronic archive is controlled through the lock and key system.

8.5.3 Refrigerator, deep freezer and desiccator are available in test item/reference item archives to ensure the different storage conditions like Refrigerator: 2°C to 8°C, Deep Freezer: -10°C to -25°C and Desiccated condition (protect from moisture).

8.5.4 Test/reference items with storage condition Ambient/RT (Room Temperature) 15°C to 30°C shall be stored in cupboards.

8.5.5 Desiccators are present in both ambient and refrigeration storage conditions and follow procedure as per **SOP-GLP-GEN-0007** (Operation, cleaning and maintenance of Desiccator).

Note: Cleaning and maintenance of desiccator should be performed only when it is in use.

8.5.6 PPE like lab coat, chemical resistant protective gloves, nose mask and chemical safety goggles shall be maintained in Test item archives for safe handling of Test/Reference/control Item.

8.6. Electronic Archives

8.6.1 A Cabinet which is fire proof and free from magnetic interference is present in

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electronic archives to store electronic records.

8.6.2 Desiccant silica gel bag should be placed in the cabinet to prevent damage of CD/DVD from moisture.

8.7. Organization and Management of Storage Facilities

8.7.1 Archive staff should ensure the repositories shall not be used to store spare furniture, equipment or consumables. However, stationery like spare empty file holders shall be maintained handy for ease of arranging documents that are archived; but this should not occupy significant space in archives.

8.7.2 The shelves used to retain records and bottled specimens/sealed polythene bags/blocks/slides stored in sealed polythene bags or packed in customized boxes should be non-combustible, non-deteriorating and rust-proof metal.

8.7.3 The archived records and materials should be arranged in appropriate order with adequate labeling as per **SOP-GLP-AC-0001** (Receipt of records, materials, procedure of archiving, allotting accession number, indexing and assigning retention period) and document the details by using a form **FORM-GLP-AC-0006** (Archival receipt-retrieval logbook).

8.7.4 Archives staff should maintain a detail inventory as data base in microsoft excel which help for easy traceability.

8.7.5 Once records and materials have been archived and placed in storage areas, they should be removed only in accordance with applicable retrieval procedures described in **SOP-GLP-AC-0003** (Procedures of retrieval and return of records and materials) and document the details by using a form **FORM-GLP-AC-0006** (Archival receipt-retrieval logbook).

8.7.6 Migration of electronic records from old to new media can be done to avoid the loss of data/space saving in archives. Refer section 8.23 for migration procedure.

8.8. Archives Activity Planner

8.8.1 Use form **FORM-GLP-AC-0015** (Archives Activity Planner) given as reference for preparing archives activity planner that the Archivist and/or Deputy shall prepare in

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the month of January (every year) and same shall be approved by TFM.

a) Planned activities within GLP archives

b) By end of the Year (as per planner), the completed activities should be updated as applicable. C-Completed as per the schedule; UP- Unplanned; UPC- Unplanned Completed; NC- Not completed. The record shall be signed, dated and archived as per **SOP-GLP-GEN-0022** (Submission of documents, electronic data and specimens to GLP archives)**.**

8.8.2 Activity planner should be updated in the month of January for previous year and/or as when required for the status of activity.

8.9. Maintenance Procedures: Storage areas should be kept in good order. The measures to be taken for proper maintenance are described in the following sections.

8.10. Temperature Recording

8.10.1 In archives temperature and relative humidity should be monitored by archivist/ staff, either using a calibrated digital thermo hygrometer or any other appropriate mode and document the details.

8.10.2 Operation and calibration procedure of thermo hygrometer should carry out as per SOP **SOP-GLP-GEN-0017** (Operation and Calibration of Digital Thermo hygrometer).

8.10.3 The staff should identify and take appropriate action when the air conditioning is not working properly inform/coordinate with EU department for necessary corrective action.

8.10.4 Archives staff should monitor the Room/ Refrigerator/ freezer temperature once daily on working days excluding holidays and declared (Preferable between 9:00 AM to 11:30 AM). Room Temperature conditions should be maintained in all designated archives

a) Room Temperature: 20-25˚C and excursion allowed up to 15-30˚C.
b) Relative Humidity: 30 to 70% and excursion allowed up to 25-85%.
c) For storage of test/reference item at recommended storage conditions, the refrigerator/freezer temperatures are maintained at 2˚C to 8˚C and -10°C to -25°C.

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8.10.5 Document the refrigerator temperature readings of the single compartment refrigerator using form **FORM-GLP-GEN-0038** (Refrigerator Temperature Record – Single Compartment) and two compartments (upper & lower compartments) using form **FORM-GLP-GEN-0039** (Refrigerator Temperature Record – Two Compartments) **(**as applicable).

8.10.6 Document the temperature readings of freezer at designated place using form **FORM-GLP-GEN-0040** (Freezer Temperature Record) Document the room temperature & Relative humidity readings using form **FORM-GLP-GEN-0037**(Room Temperature and Relative Humidity Record).

8.10.7 If the readings are not observed in the specified range then intimate the same to equipment custodian and/or EU personnel for necessary appropriate corrective actions. Report the breakdown as per procedure in SOP **SOP-GLP-EU-0015** (Preventive Maintenance, Calibration and Breakdown Maintenance by using SAP application).

8.11. Preventive Maintenance

8.11.1 Engineering & Utility (EU) department personnel/external vendor are responsible for conducting preventative maintenance in consultation with the equipment custodian as per form **FORM-GLP-EU-0020** (Preventive Maintenance Task list).

8.11.2 PM should be carried out once a year by qualified service technicians and verified by equipment custodian and/ or EU personnel.

8.12. Validation of Refrigerator/Freezer

8.12.1 **Duration:** Validation of refrigerator/freezer should be carried out once in a year and/or as and when required.

8.12.2 **Condition:** Validation activity should be carried out in empty/unloaded condition.

8.12.3 Before initiating the validation/PM activity the test items should shift to another location such as any stand by refrigerator/freezer/dry ice box/TICO office.

Note 1: *If the test items are shifted to TICO office, the test item stored boxes should seal with sign & date of archivist/deputy and details should.*

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Note 2: *The transfer details of test items along with location & equipment Id should document in the respective inventory logbooks and follow the entry exit procedure as per SOP* ***SOP-GLP-TC-0001*** *(Access, Entry and Exit Procedure in Test Item Control Office).*

8.12.4 After completion of the calibration, equipment should be labelled with new calibration status and test items should shift back to the respective location.

8.12.5 In refrigerator/freezer, sensor should be placed at hot spot location as identified in validation/calibration report.

8.12.6 Breakdown management and contingency plan flow the procedure describe in the SOP **SOP-GLP-GEN-0019** (Inventory Management, Cleaning and Maintenance of Refrigerators/Freezers).

8.13. Disposal

8.13.1 If the device becomes no longer fit for purpose, it must be disposed as per SOP No: **SOP-GLP-EU-0012** **(**Retirement and Disposal Procedure for GLP Instrument/Equipment including Computerized Systems).

8.14. Lighting

8.14.1 Fire-proof lightening system is provided to minimize the risk of electric short circuit.

8.14.2 Archives staff should ensure sufficient lighting in the archives. If lights are damaged inform the EU department and get it replaced.

8.14.3 It is ideal and preferable that the lights should be switched on while working and switched off when work is completed to avoid damage to the archives and unnecessary use of electricity.

8.15. Housekeeping procedures

8.15.1 Housekeeping activity should be carried out preferably once in a week on any one business working day. Housekeeping staff enters archives for routine cleaning activities such as brooming, dusting, moping etc., accompanied by the archives staff. The activity shall be monitored by archives staff and record details using form **FORM-GLP-AC-0017** (Housekeeping Activities – Archives).

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8.16. Pest and Rodent Control

8.16.1 To prevent any damage by insects and/or pest to archives, storage areas shall be subjected to pest control activity. This should be facilitated by SH&E department. Frequency and procedure of Pest control activity shall be carried out preferable once in 2 months.

8.16.2 Pest control personnel should be accompanied by the archives staff.

8.16.3 Rodent bait should be placed near the repositories.

8.16.4 Archives staff should be documenting all activities performed using respective logbooks and follow the procedure described as per form **FORM-GLP-AC-0013** (Entry and Exit logbook- Archives).

8.17. Verification of archives

8.17.1 Periodic verification of archived documents and materials should be performed to ensure the quality and integrity of archived documents and materials.

8.17.2 Archives staff should conduct an annual verification of the storage shelves/locations and physical condition of the archived records/materials in Document/Specimen/Test Item/Reference Item Archives in the month of **Jan/ Feb** to assess the following

a. Signs of mould growth

b. Signs of insect infestation

c. Labels missing from shelves or boxes and storage containers and legibility of details mentioned on label.

d. Shelves or other equipment in need of repair or replacement

e. Leakage/breakage of any containers used to store test/reference item

8.17.3 During verification if any discrepancies observed take an appropriate action to resolve the discrepancies and take help from other departments wherever required.

8.17.4 Document the details of verification, discrepancies observed and action taken to resolve the discrepancy in form **FORM-GLP-AC-0016** (Annual Verification of Archives).

8.17.5 To ensure the integrity of archived electronic data, CD/DVDs shall be verified annually once. For verification of CD/DVDs follow the procedure described in section **8.22.**

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8.18. Checking Formalin Levels in Specimen Archives

8.18.1 The adequate level of formalin should maintain in the wet tissue storage container or package to maintain the intensity of wet tissues for log duration. To fulfil this requirement formalin level in storage container shall be verified yearly twice.

8.18.2 Specimens should be stored in 10% Neutral Buffered Formalin (NBF). Specimens should submit for archival in bottles which are marked with permanent marker indicating the ideal formalin level (3/4 bottle) or in sealed polythene bags leakage of formalin shall be checked.

8.18.3 Archives staff shall check the formalin levels of bottled specimens/specimen in sealed polythene bags in the month of **May/Jun** & **Nov/Dec** in a year.

8.18.4 Wear appropriate PPE (lab coat, chemical resistant protective gloves, nose mask and chemical safety goggles).

8.18.5 If 10% NBF levels are below the mark; archives staff shall request pathology department, to provide 10% NBF to refill specimen bottles, provided formalin should be stored in specimen archives Archivist/Staff should complete the refilling activity in specimen archives. On completion of refilling activity, make an entry using form **FORM-GLP-AC-0018** (Verification of formalin level and repacking requirement in specimen archives). Ensure to close the bottles tightly after filling.

8.18.6 If archivist feels bottles are getting damaged; then take an expert opinion from the Pathology team.

8.18.7 Identify bottled specimen/polythene bag, which needs to be re-packed/changed. Pathology team should prepare appropriate labels.

8.18.8 Archivist/Deputy should inform Management and/or Head-QA about the materials which required repacking/change of bottles/polythene bags via mail with attached list compiled in excel sheet. No, formal retrieval request is required, but the details of transfer shall document by using a form **FORM-GLP-AC-0019 (**Repacking of Specimens).

8.18.9 Transfer materials using a trolley to necropsy bay at Toxicology facility. Transfer contents into labeled bottles/sealed covers using Laminar Air Flow cabinet with the

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help of pathology team into bottles/cover and seal them.

8.18.10 After completing the above activity received materials details should document by using a form **FORM-GLP-AC-0019 (**Repacking of Specimens).

8.18.11 **Minor Formalin Spill (Leakage from bottled specimen/sealed polythene bag)**

8.18.11.1 Specimen bottles have tight fitting lids. Specimens in polythene bags should be packed using 02 polythene covers and sealed with sealer.

8.18.11.2 Minor spillage may happen from leaky specimen container/sealed polythene bag due to a poorly secured lid/sealer.

8.18.11.3 A minor spill is capable of being cleaned up without any assistance and does not require an emergency response.

8.18.11.4 Wear appropriate personal protective equipment such as lab coat, gloves, nose mask and safety goggles.

8.18.11.5 Absorb the spilled material using paper towels provided in the spillage kit.

8.18.11.6 Place paper towels into a waste bag and seal to minimize the release of formaldehyde vapours. Do not dispose into a general waste bin. Inform housekeeping staff. Categorize waste as per annexure **WI-GLP-SHE-0002** (Categories of Biomedical waste).

8.18.11.7 Ensure contaminated area is washed twice with soap and water by housekeeping staff under the supervision.

8.18.12 **Moderate Formalin Spill**

8.18.12.1 Alert SH&E department using intercom facility or via mobile phone.

8.18.12.2 Wear appropriate PPE protective clothing, gloves, safety goggles, shoe covers and an appropriate nose mask.

8.18.12.3 Quickly block or contain size and spread of spill by using absorbent spill control kit. Encircle the spill perimeter using the spill control kit. Mix absorbent into spill working from the outside in. If formalin neutralizing agents are used, wait 5 - 10 minutes until the reaction stops. Inform housekeeping staff.

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8.18.12.4 Scrape up absorbed spill residue with a plastic pan or remove pillows and bunds. Place material in a waste bag. Seal the bag tightly to minimize the release of formaldehyde fumes. Categories waste as per annexure WI-GLP-SHE-0002 (Categories of Biomedical waste). Dispose as contaminated waste as per procedure in SOP SOP-GLP-SHE-0004 (Bio Waste Management).

8.18.12.5 Ensure contaminated area is washed twice with soap and water by housekeeping staff under the supervision.

8.19. First Aid Measures

8.19.1 **Eyes:** Flush eyes with plenty of water for at least 15 minutes, occasionally lifting the upper and lower eyelids. Do not allow person to rub eyes or keep eyes closed. Use the eyewash bottle provided.

8.19.2 **Skin:** Immediately flush skin with plenty of water for at least 15 minutes while removing contaminated clothing and shoes. Wash clothing before reuse. Destroy contaminated shoes. Get medical aid if irritation develops or persists.

8.19.3 **Ingestion:** If person is conscious and alert, give 2-4 cupsful of milk or water. Never give anything by mouth to an unconscious person. Get medical aid immediately.

8.19.4 **Inhalation:** Get medical aid immediately. Remove from exposure and move to fresh air area immediately.

8.19.5 Water in eyewash bottle shall be changed once in 15 business working day by archives staff and document the next due date (use any label template to update the details).

8.20. Periodic Verification of CD/DVD

8.20.1 Activity should be carried out in the month of **Jul/Aug** in a year by the Archivist/Deputy with the help of IT department.

8.20.2 About 30% of the CD/DVD should be verified randomly during periodic verification.

8.20.3 Verification of CD/DVD should be documented using logbook **FORM-GLP-AC-0014 (Verification of CD/DVD in electronic archives).**

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8.20.4 During verification if any discrepancies noticed/observed should informed to IT personnel and get resolved, document the action taken details in remarks column using **FORM-GLP-AC-0014 (Verification of CD/DVD in electronic archives)**.

8.20.5 If noticed discrepancies cannot be solved inform the management & head QA for action to be taken. Management is responsible for delegate decision to be taken.

8.21. Migration of Electronic Data

8.21.1 Migration of archived electronic data should perform in specific conditions, suspect of data loss, space saving in electronic archives.

8.21.2 Enlist the details of records to be migrated and communicate to Head QA & Management.

8.21.3 Management is responsible for making a decision about transfer of records.

8.21.4 Upon approval archivist staff with the help of IT personnel shall initiate migration process of e-records and document the details using form FORM-GLP-AC-0020 (Migration of Electronic Data).

8.21.5 After completion of migration QA personnel shall verify the migrated data for completeness.

8.21.6 The new generated media should archive with all details by archivist/staff the same details should be updated in respective data base. In data base new accession number should update along with old.

Note: Due to any unavoidable emergency situations if Archivist/Archives Staff is not available in office premises during working days. The routine activities like temperature monitoring/recording, Housekeeping, and water change in eye wash bottle, PM of Smoke detector, Pest control activity, changing of silica gel in desiccator shall not be performed. The activities shall be planned and performed once resume to work.

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9.0 REFERENCE(S):

Document No.	Title of the Document
SOP-GLP-AC-0001	Receipt of Records, Materials to Archives, Procedures for archiving, allotting accession number, indexing and assigning retention period.
SOP-GLP-AC-0003	Procedures for Retrieval and Return of Records and Materials
SOP-GLP-AC-0004	Access, Entry & Exit Procedures in GLP Archives
SOP-GLP-EU-0010	Routine Maintenance and Validation of Laminar Air Flow Unit (s)
SOP-GLP-EU-0012	Retirement and Disposal Procedure for GLP Instrument/Equipment including Computerized Systems
SOP-GLP-EU-0015	Preventive Maintenance, Calibration and Breakdown Maintenance by using SAP application
SOP-GLP-GEN-0007	Operation, Cleaning and Maintenance of Desiccator
SOP-GLP-GEN-0017	Operation and Calibration of Digital Thermo hygrometer
SOP-GLP-GEN-0022	Submission of Documents, Electronic Data and Specimens to GLP Archives
SOP-GLP-TC-0001	Access, Entry and Exit Procedure in Test Item Control Office
SOP-GLP-SHE-0004	Bio Waste Management

10.0 ANNEXURE(S):

Document No.	Title of the Document
FORM-GLP-AC-0013	Entry and Exit logbook- Archives
FORM-GLP-AC-0014	Verification of CD/DVD in electronic archives
FORM-GLP-AC-0015	Archives Activity Planner*
FORM-GLP-AC-0016	Annual Verification of Archives
FORM-GLP-AC-0017	Housekeeping Activities - Archives
FORM-GLP-AC-0018	Verification of formalin level and repacking requirement in specimen archives
FORM-GLP-AC-0019	Repacking of Specimens
FORM-GLP-AC-0020	Migration of Electronic Data *
FORM-GLP-EU-0020	Preventive Maintenance Task list

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FORM-GLP-GEN-0037	Room Temperature and Relative Humidity Record
FORM-GLP-GEN-0038	Refrigerator Temperature Record – Single Compartment
FORM-GLP-GEN-0039	Refrigerator Temperature Record – Two Compartments
FORM-GLP-GEN-0040	Freezer Temperature Record
WI-GLP-SHE-0002*	Categories of Biomedical waste

*Editable forms.

11.0 CHANGE HISTORY

Quality Record No	Brief details of changes
CC1000087018	Converted to current SOP template. Amendment details are added and included procedure for preventive maintenance and validation of refrigerator/freezer. Forms (FORM-GLP-AC-0014, FORM-GLP-AC-0015, FORM-GLP-AC-0016, FORM-GLP-AC-0017, FORM-GLP-AC-0018, FORM-GLP-AC-0020) are updated in Dochub template. In form FORM-GLP-AC-0019 minor modification has done for better clarity & compliance.

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Intention	Employee ID	User Name	Date (IST) DD/MM/YYYY
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Approved By Test Facility Management	P00068654	D Ravi Krishna Babu.	12/08/2022 17:43:40

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